

PAPER_1588 — Coulomb constant $k_e \approx k_e = N_{CH} - F \cdot SSQ + F^2 \cdot D_{phys} = 8.988$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Electromagnetism **Date:** June 18, 2026 **Status:** CLOSED — 0.056% residual near-EXACT closure

Observation

PAPER_1209DD_S617 (Electromagnetism Unified Proof Set) gives the lead-digit form:

Coulomb constant $k_e \approx k_e = N_{CH} - F \cdot SSQ + F^2 \cdot D_{phys} = 8.988$

Residual to CODATA: **0.056%**.

UQFF Closed Identity

$k_e = N_{CH} - F \cdot SSQ + F^2 \cdot D_{phys} \approx 8.988$ residual 0.056%

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** (ϵ_0, μ_B, k_e) cluster around **$K_{MEX} \cdot D_{phys} = 8.33$** or **$N_{CH} = 9$** - **Planck/quantum constants** ($\hbar$) cluster around **$D_{BSFG} = 6$** - **Relativistic constants** (c) cluster around **$SO_5/D_{phys} = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Coulomb constant k_e as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S617
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "coulomb_ke_lead_8_988"})`

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